

UNIT 2 – GAUSS’S LAW

2.3 Gauss’s Law: Cylindrical and Planar Symmetry**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. An infinitely long line of charge has linear density $\lambda = 2.0 \mu\text{C}/\text{m}$. Find E at a perpendicular distance of 0.050 m.

Gaussian cylinder of radius r and length ℓ : $E(2\pi r\ell) = \lambda\ell/\varepsilon_0$, so $E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r} = \frac{2(8.99 \times 10^9)(2.0 \times 10^{-6})}{0.050} = 7.2 \times 10^5 \text{ N/C}$.

2. A large flat insulating sheet carries surface charge density $\sigma = 1.0 \times 10^{-6} \text{ C}/\text{m}^2$. Find the field near the sheet.

Gaussian pillbox with faces of area A on both sides: $2EA = \sigma A/\varepsilon_0$, so $E = \frac{\sigma}{2\varepsilon_0} = \frac{1.0 \times 10^{-6}}{2(8.85 \times 10^{-12})} = 5.6 \times 10^4 \text{ N/C}$, perpendicular to the sheet.

3. Explain why the field just outside a charged conductor in electrostatic equilibrium is $E = \sigma/\varepsilon_0$, twice the value for the thin sheet above.

Inside a conductor in equilibrium $E = 0$, so a pillbox straddling the surface has flux only through its outer face: $EA = \sigma A/\varepsilon_0$, giving $E = \sigma/\varepsilon_0$. The thin sheet has flux through both faces, which halves the result.